

Experiment-1 write the python program to solve 8 puzzle problem

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EXPERIMENT-1.py - C:/Users/prana/OneDrive/Desktop/EXPERIMENT-1.py (3.13.14)
File Edit Format Run Options Window Help

start = [1, 2, 3,
         4, 0, 6,
         7, 5, 8]

goal = [1, 2, 3,
        4, 5, 6,
        7, 8, 0]

moves = 0

print("Initial State:")
print(start[0:3])
print(start[3:6])
print(start[6:9])

start[4], start[7] = start[7], start[4]
moves += 1

print("\nMove", moves)
print(start[0:3])
print(start[3:6])
print(start[6:9])
|
start[7], start[8] = start[8], start[7]
moves += 1

print("\nMove", moves)
print(start[0:3])
print(start[3:6])
print(start[6:9])

if start == goal:
    print("\nPuzzle Solved!")
    print("Total Moves =", moves)
else:
    print("\nPuzzle Not Solved")
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IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> ===== RESTART: C:/Users/prana/OneDrive/Desktop/EXPERIMENT-1.py =====
Initial State:
[1, 2, 3]
[4, 0, 6]
[7, 5, 8]

Move 1
[1, 2, 3]
[4, 5, 6]
[7, 0, 8]

Move 2
[1, 2, 3]
[4, 5, 6]
[7, 8, 0]

Puzzle Solved!
Total Moves = 2
>>> |
```